

Exceptional Expertise to Lung Health



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In India, with a diverse composition of gene pool, regional habitats, food habits and social beliefs, where are pulmonary diseases more common and why? And of what kind are they?

In the population, respiratory diseases are mainly related to exposures like smoking, air pollution and infections in specific regions and influenced by social beliefs rather than directly affected by genetics. The most common respiratory diseases in India are, as anywhere else in the world, Asthma, Chronic

obstructive pulmonary disease (COPD/emphysema/ chronic bronchitis), Pulmonary Tuberculosis, bronchiectasis and now Covid-19 pneumonia. In geographical terms, the Gangetic plains of north India have high levels of air pollution, especially in winters, as compared to anywhere else in the country and the world. 21 out of the top 30 cities with the worst air quality in the world are in India and all of them are in the Gangetic belt. It is estimated that air pollution causes an excess of 20,00,000 (twenty lakh) deaths in our country alone each year. Air pollution, both indoors and outdoors and its direct and indirect effects cannot be ignored. Apart from outdoor air pollution related to industries, vehicular exhaust, construction and stubble burning, indoor air pollution is also very important and is related to wood stoves, kerosene stoves and other biomass fuels for heating in poorly ventilated rooms. This is relevant especially in the winter season for North India and disproportionately affects low socio-economic groups. It causes poor peak lung health and recurrent childhood respiratory infections in children and leads to development of COPD and pulmonary infections in adults. It is associated with an excess of respiratory and cardiovascular deaths in the community. There is a silent epidemic of lung cancers in non-smokers related to air pollution in our country and this is especially pronounced in the Gangetic belt. Allergic diseases also tend to cluster more in urban areas than rural areas due to interaction of allergens with diesel vehicular exhaust and lack of endotoxin exposures in childhood. The impact of today's low-grade toxic environmental exposures on lung function will be

delayed and tends to show up 20-25 years later; similarly, any beneficial effect of legislation will take two decades to fructify and the time to act is now.

Environmental pollution is a major cause of pulmonary diseases. What is the impact of such constant exposure to bad air on the lungs and how can one prevent the lungs turning ill?

You are what you breathe and what you eat. Exposure to bad quality air is as bad as smoking a cigarette. We breathe 11000 litres of air in a day and so, the effects will be cumulative over time and will be seen in terms of worsening respiratory health, frequent infections, excess of heart attacks and strokes and several cancers. It is a fundamental right of every citizen to have good air to breathe. Societal changes, political will and legislation are needed for effective and large-scale changes. At a personal level, one can contribute by lowering one's contribution to air pollution by using public transport/car-pooling etc whenever possible and by spreading awareness of the ill effects of air pollution. People with respiratory diseases must be aware of the quality of air, particularly during change of season and festivals. Stay indoors during these times and wear an N95 mask and restrict duration of exposure if you are compelled to go out for any errand. Get yourself vaccinated for Pneumococcus and annually for Influenza apart from COVID-19 and pay attention to hand hygiene, cough etiquette and self-isolation when you have lung infection to prevent it from spreading to others.

What are the conditions under which the pulmonary system gets compromised and how do you identify those symptoms?

